PHASE B - ABSTRACTION CYCLE 2

Phase B - Local-to-Global Polynomial Image Compression

How sparse polynomial images modulo completely split primes, combined with Minkowski height control, can force a small global image without global rank-one factorization.

Cycle verdict
The mechanism is rigorous and reusable once one adds a uniform per-residue-class packing bound. The decisive inequality is $2a-c+s>0$. Pohoata's quadratic construction satisfies it with $\theta=13D\rho<1$. Local congruence compression therefore gives a genuine route around global factorization failure, but candidate supply and removable arithmetic deformation remain the bottleneck.

Scope discipline

This report runs exactly investigations B6-B10. It does not modify Phase A, either frozen unit-distance guide, or the Cycle 1 PDF. It does not claim novelty and does not begin Phase C.

Independent study notes - 26 August 2026

How to read this report

Epistemic labels

Label	Meaning
SOURCE	Directly stated or proved in a verified supplied project source.
BACKGROUND	Standard mathematics or an elementary derivation.
ABSTRACTED PRINCIPLE	A quantitative principle proved at the stated level from verified mathematics.
TRANSFER HYPOTHESIS	A proposed extension not yet established.
EXTERNAL RESULT	A theorem verified in literature outside the supplied project sources.
FAILURE	A false, insufficient, or quantitatively useless proposed principle.
OPEN	A precise unresolved mathematical question.

Verified source hierarchy

Tier	Source	Use
Principal external	C. Pohoata, Split primes and the Elekes-Ronyai problem, arXiv:2606.13619v1	All polynomial construction claims; equations and proposition numbers follow the paper.
Authoritative context	Elekes-Ronyai (2000), checked through the journal record and a later primary research restatement	Precise additive and multiplicative exceptional forms and qualitative theorem.
Project source	Planar Point Sets with Many Unit Distances and human-verified Remarks	Tower, split-prime, window, and rate comparison.
Project source	Frozen Phase A and Cycle 1	Minkowski box estimates, global mechanisms, and the prior bottleneck.
Discovery only	unit-distance-cot.pdf	Not used as proof authority.

The five investigations

ID	Investigation	Audit outcome
B6	Reproduce the split-prime polynomial mechanism	VERIFIED exactly as normalized in the protocol
B7	Combine several split primes	VALID CRT product density ρ^d
B8	Convert local sparsity to global support	VALID through per-class Minkowski packing
B9	Abstract the local-to-global lemma	VALID after adding a uniformity hypothesis
B10	Degree-versus-local-density analysis	VALID signed threshold; degree two is favorable

1. Executive overview

Cycle 1 asked whether a polynomial globally factors through an additive or multiplicative one-dimensional operation. Cycle 2 asks a different question: can a polynomial be globally nonspecial but locally sparse modulo many arithmetic coordinates?

local restriction -> split residue coordinates -> density ρ^d -> per-class packing -> full support bound.

[EXTERNAL RESULT] Pohoata proves that the fixed quadratic polynomial

$$f(x,y)=x+y+(x-y)^2$$

has arbitrarily large finite real sets A satisfying $|f(A,A)|\leq|A|^{2-c}$ for some absolute $c>0$, even though f is neither additive nor multiplicative in the Elekes-Ronyai sense.

The main inequality

Let a be the carrier exponent, B an upper size exponent, c the candidate output exponent before the local saving, and s the local sparsity exponent. Then

$$|F(A_d,A_d)| \leq e^{(c-s)d+o(d)},\ |A_d|^2 \geq e^{2ad-o(d)}.$$

The normalized saving is positive precisely when

$$\delta = 2a-c+s > 0.$$

If also $|A_d|\leq e^{Bd+o(d)}$, then

$$|F(A_d,A_d)| \leq |A_d|^{2-\delta/B+o(1)}.$$

What this means
Local sparsity is not merely collision evidence. Once every allowed congruence class contains only a controlled number of points in the archimedean box, it directly bounds every possible output. This is a full-support certificate.

2. Why Cycle 1 was insufficient

Cycle 1 correctly separated collision diagnostics from support proofs and identified exact factor-through compression. Its first genuinely mixed test, $h(x,y)=xy(x+y)$, resisted the global $A=GP$ mechanism.

[SOURCE] Cycle 1 showed that for $A=\{q^0,\dots,q^{N-1}\}$ with $q\geq 3$,

$$|h(A,A)|=N(N+1)/2,\ E_h(A)=2N^2-N.$$

Thus strong multiplicative structure can coexist with an essentially quadratic image. The obstruction is the rank-two expression

$$h(g_1p_1,g_2p_2)=g_1^2g_2p_1^2p_2+g_1g_2^2p_1p_2^2.$$

A new route

Local-to-global compression does not try to separate these real variables globally. It asks whether the reduced polynomial image occupies only a strict subset of a finite residue field and whether complete splitting repeats that restriction d times.

[ABSTRACTED PRINCIPLE] Global rank-one factorization and local residue sparsity are logically different sufficient mechanisms. Neither class is claimed to exhaust polynomial compression.

3. The new local-to-global mechanism

One fixed rational prime can create many algebraic coordinates if it splits completely. The arrows below are algebraic identities and counting steps, not a probabilistic independence heuristic.

Arithmetic input	Complete splitting	CRT coordinates	Reduced image	Amplified density
One odd prime p	$pO_K=p_1\cdots p_d$	$O_K/pO_K\cong F_p^d$	d square restrictions	density q_p^d

For Pohoata's polynomial, $q_p=(p+1)/(2p)$. Several rational primes multiply their densities:

$$\rho = \Pi_{p|Q}\,(p+1)/(2p), \text{ allowed density modulo } QO_K = \rho^d.$$

Why two controls are required

Constraint	What it gives	Why it is insufficient alone
Archimedean	Outputs lie in a finite Minkowski box	The box may still contain essentially $\|A\|^2$ points
Non-archimedean	Outputs lie in few residue classes	Each allowed class contains infinitely many algebraic integers
Both	Few classes times a uniform maximum per class	This directly controls the full support

4. Proof-level reproduction of Pohoata's construction

We first state the exact source ingredients, then derive the support estimate in Sections 5-8. This prevents the abstraction from drifting away from the verified construction.

[EXTERNAL RESULT] Pohoata uses

$$f_Q(x,y)=Q(x+y)+(x-y)^2,\ Q=p_1\cdots p_t,$$

where the p_i are distinct odd rational primes. This exactly matches the normalization proposed in the Cycle 2 protocol.

[EXTERNAL RESULT] There are totally real fields K of degrees d tending to infinity, a constant D, and an infinite fixed set of odd rational primes splitting completely in every field, with $\Delta_K^{1/d}\le D$.

Choose finitely many primes from that split set so that

$$\rho=\Pi_{i=1}^{t}(p_i+1)/(2p_i),\ \theta=13D\rho<1.$$

After these primes are chosen, Q, ρ, and θ remain fixed. Choose

$$X \ge \max\{Q, 2D^{1/2}, 1\},\ P=B_K(X),\ A_0=\sigma_1(P).$$

Here σ_1 is one real field embedding. It is injective, so $|A_0|=|P|$ and it preserves polynomial identities with rational coefficients.

Source output

$$|f_Q(P,P)|\le\theta^d|P|^2,\ \theta<1.$$

Since $|P|\le(2X+1)^d$, the paper obtains the explicit saving exponent

$$c = -\log\theta / \log(2X+1) > 0.$$

What this means
The parameter sent to infinity is the field degree d. The primes, Q, the box radius X, and every exponential base are fixed first. This is essential for producing one power saving rather than a changing-polynomial statement.

5. B6 - Local residue restriction

The polynomial is engineered so that its linear term disappears modulo every prime dividing Q. What remains is a square, whose exact finite-field image is easy to count.

Reduction modulo a prime ideal

Fix p|Q and a prime ideal pfrak above p. Since Q belongs to pfrak,

$$f_Q(x,y)=Q(x+y)+(x-y)^2 \equiv (x-y)^2 \pmod{\text{pfrak}}.$$

If p splits completely in a degree-d field K, then

$$pO_K=\text{pfrak}_1\ldots\text{pfrak}_d,\ O_K/\text{pfrak}_j\cong F_p.$$

[BACKGROUND] Complete splitting means there are d distinct unramified prime ideals, each with residue degree one. Hence every quotient has exactly p elements.

Exact size of the square set

The map $z\to z^2$ on F_p^* has kernel {1,-1}, so it has (p-1)/2 nonzero values. Including zero gives

$$|\{z^2:z \text{ in } F_p\}|=(p+1)/2,\ q_p=(p+1)/(2p).$$

For every odd p, $q_p \le 2/3<1$.

REPRODUCE -> ABSTRACT -> AUDIT

[ABSTRACTED PRINCIPLE] A useful local restriction is an exact bound on the reduced value set, not a statement that outputs collide often.

[FAILURE] Calling the d restrictions independent is legitimate only through the product decomposition of quotient rings. No random variables are present.

What this means
Each residue coordinate permanently removes the same positive fraction of possible outputs. Complete splitting makes the number of such coordinates grow with d.

6. B7 - Several split primes and CRT amplification

One split prime repeats one local restriction d times. Several fixed split primes stack different restrictions without conflict because all corresponding prime ideals are pairwise comaximal.

Chinese remainder decomposition

$$O_K/QO_K \cong \Pi_{i=1}^{t} \Pi_{j=1}^{d} O_K/\mathfrak{p}_{i,j} \cong \Pi_{i=1}^{t} F_{p_i}^{d}.$$

The full quotient has Q^d classes. In the coordinate belonging to p_i, f_Q can use only $(p_i+1)/2$ square classes. Thus the allowed class set Ω satisfies

$$|\Omega| \le \Pi_{i=1}^{t}((p_i+1)/2)^d = (\rho Q)^d.$$

Dividing by Q^d gives the exact ambient density bound

$$|\Omega|/|O_K/QO_K| \le \rho^d,\ \rho=\Pi_{i=1}^{t}(p_i+1)/(2p_i)<1.$$

Why complete splitting matters

If p has only g prime ideals above it, there are only g residue coordinates. An inert prime has g=1, so its local loss stays a fixed constant rather than becoming q_p^d. Complete splitting is the case g=d and gives the maximal degree exponent.

What this means

The gain is s=-log ρ per field degree. Adding fixed primes adds their gains. The cost is that Q and the coefficients of f_Q grow, so those terms must be audited before the primes are accepted.

7. The archimedean Minkowski carrier

The local restrictions live in finite quotient rings. To turn them into a finite real set, the construction cuts out a large box in the ring of integers and later projects it through one real embedding.

Definition and lattice geometry

$$B_K(X)=\{\alpha \text{ in } O_K: |\sigma_i(\alpha)|\le X \text{ for } 1\le i\le d\}.$$

The Minkowski embedding sends O_K to a full-rank lattice in R^d of covolume $\Delta_K^{1/2}$. Blichfeldt's lemma and unit separation give

$$X^d\Delta_K^{-1/2} \le |B_K(X)| \le (2X+1)^d.$$

Under $\Delta_K^{1/d}\le D$, if $n=|B_K(X)|$ then

$$(X/D^{1/2})^d \le n \le (2X+1)^d.$$

Uniform packing in one residue class

If α and β are distinct but congruent modulo RO_K, then $\alpha-\beta=R\gamma$ for a nonzero algebraic integer γ. Some embedding satisfies $|\sigma_i(\gamma)|\ge 1$, so α and β are R-separated in the sup norm.

Consequently a radius-M box contains at most

$$(2M/R+1)^d$$

points from any one residue class modulo RO_K. This is Pohoata's Lemma 3.3.

What this means
The discriminant controls how large the input carrier is. Norm separation, not equidistribution, controls how many outputs can sit in each allowed congruence class.

8. B8 - From local sparsity to full global support

We now combine the two constraints. This is the load-bearing step where a local congruence fact becomes an upper bound on every global output.

Archimedean output radius

For a,b in $P=B_K(X)$, every real embedding satisfies

$$|\sigma_i(f_Q(a,b))| \le 2QX+4X^2 \le 6X^2,$$

because $X \ge Q$. Therefore $f_Q(P,P)$ lies in $B_K(6X^2)$.

Allowed classes times points per class

There are at most $(\rho Q)^d$ allowed classes. With $R=Q$ and $M=6X^2$, each class contributes at most $(12X^2/Q+1)^d$ points. Hence

$$|f_Q(P,P)| \le (\rho Q)^d(12X^2/Q+1)^d$$
$$= [\rho(12X^2+Q)]^d \le (13\rho X^2)^d.$$

The carrier lower bound implies $X^{2d} \le D^d n^2$. Therefore

$$|f_Q(P,P)| \le (13D\rho)^d n^2 = \theta^d n^2.$$

[EXTERNAL RESULT] Pohoata chooses the fixed split primes so that $\theta=13D\rho<1$. Since $n \le (2X+1)^d$, this gives $|f_Q(P,P)| \le n^{2-c}$ with $c=-\log\theta/\log(2X+1)$.

RATE AUDIT

Entry	Per-degree contribution	Role
Local squares	log ρ<0	Gain
Residue-class packing	log 13	Cost
Carrier discriminant	log D	Cost
Net normalized rate	log θ=log(13Dρ)<0	Saving
Parameter to infinity	d	Amplifier

What this means
Unlike collision energy, this count sees every output: each one has an allowed residue label and lies in the bounded box. There is no uncontrolled light tail.

9. B9 - General local-to-global compression lemma

The paper's proof suggests a reusable theorem, but the schematic version needs one extra hypothesis: a uniform count inside each residue class.

Formal lemma

[ABSTRACTED PRINCIPLE] Let K_d be degree-d number fields, A_d finite carriers, and R a fixed positive integer. Assume:

- Carrier rates: $e^{ad-o(d)} \le |A_d| \le e^{Bd+o(d)}$, with $0<a\le B$.
- Local support: outputs occupy Ω_d modulo RO_{K_d}, with $|\Omega_d| \le e^{(\log R-s)d+o(d)}$.
- Uniform class packing: each residue class contributes at most $e^{ud+o(d)}$ points from the output region.
- Every projection, scaling, and parameterization has its stated bounded fiber cost already included in a, B, u, or s.

Set $c=\log R+u$. Then

$$|F(A_d,A_d)| \le e^{(c-s)d+o(d)}.$$

Comparison with the carrier gives

$$|F(A_d,A_d)|/|A_d|^2 \le e^{-(2a-c+s)d+o(d)}.$$

Thus the exact signed condition is

$$\delta:=2a-c+s>0 \text{ equivalently } s>c-2a.$$

Using the upper size rate B converts this to

$$|F(A_d,A_d)| \le |A_d|^{2-\delta/B+o(1)}.$$

Adversarial correction

[FAILURE] If one knows only that the entire archimedean region contains e^{cd} points and that the allowed residue set has density e^{-sd}, the bound $e^{(c-s)d}$ need not follow. All region points could lie in the allowed classes.

What this means
The strongest Cycle 2 abstraction is not 'local density times global volume.' It is 'number of allowed classes times a worst-case bound for one class,' followed by a signed rate comparison.

10. B10 - Degree versus local density

Cycle 1 found that generic polynomial height can consume the quadratic output budget. Local sparsity can compensate, but the exact threshold depends on degree, coefficient height, modulus, and discriminant.

Generic archimedean scale

Let $F(x,y)=\Sigma c_{ij}x^iy^j$ have total degree m and $H=\Sigma|c_{ij}|$. For $X\geq 1$ and a,b in $B_K(X)$,

$$|\sigma_k(F(a,b))| \leq H\,X^m = M.$$

Suppose the image modulo RO_K uses at most $(\rho R)^d$ classes. Uniform packing gives

$$|F(A,A)| \leq [\rho(2HX^m+R)]^d.$$

Since $|A|^2\geq(X^2/D)^d$,

$$|F(A,A)|/|A|^2 \leq \Theta^d,$$

$$\Theta = \rho D(2H\,X^{m-2} + R/X^2).$$

The reusable test is $\Theta<1$. The resulting power-saving exponent is at least $-\log\Theta/\log(2X+1)$.

Why degree two is favorable

Degree	Main scale after normalization	Effect of increasing X
m<2	X^{m-2} decays	Helps the ledger
m=2	$X^0=1$	Does not enlarge the main degree cost
m>2	X^{m-2} grows	Makes the required local density smaller

For m>2, local sparsity must satisfy the explicit threshold

$$\rho < 1 / [D(2H\,X^{m-2}+R/X^2)].$$

Coefficient and modulus audit

Adding primes can shrink ρ, but it also enlarges R and may enlarge H. These changes must be inserted jointly. Pohoata avoids the crude $H\approx Q$ cost by using the sharper shape-specific estimate $2QX+4X^2\leq 6X^2$ after choosing $X\geq Q$.

What this means
Degree two matches the X^{2d} size of the two-input carrier. Local density then only has to beat fixed arithmetic constants. Higher degree leaves an extra $X^{(m-2)d}$ cost that the local restriction must also pay.

11. Fixed-polynomial rescaling

The congruence device f_Q contains Q, but the theorem must concern one polynomial while the sets grow. The paper removes Q exactly, with no fiber loss.

[EXTERNAL RESULT] Define $f(x,y)=x+y+(x-y)^2$. Then

$$f_Q(Qx,Qy)=Q^2f(x,y).$$

Given A_0, set $A=Q^{-1}A_0$. Then

$$f(A,A)=Q^{-2}f_Q(A_0,A_0).$$

Multiplication by a nonzero scalar is a bijection on R. Therefore

$$|A|=|A_0|,\ |f(A,A)|=|f_Q(A_0,A_0)|.$$

The input and output fiber multiplicities are both exactly one. Additive doubling is also preserved.

General criterion

[ABSTRACTED PRINCIPLE] If bijective affine maps τ_Q,η_Q satisfy $F_Q(\tau_Q(x),\tau_Q(y))=\eta_Q(F(x,y))$, then $A=\tau_Q^{-1}(A_0)$ preserves both input and output cardinalities. Bounded-fiber nonbijective variants must charge those fibers explicitly.

What this means

The sparse local image belongs to an integral model f_Q, not directly to the final fixed f. The rescaling transports the support estimate exactly, so the auxiliary arithmetic parameter disappears from the theorem statement.

12. Elekes-Ronyai context and nonspecialness

The local construction matters only because the final polynomial lies outside the known global one-dimensional obstructions.

[EXTERNAL RESULT] For a fixed real bivariate polynomial f, Elekes and Ronyai proved the following qualitative statement: if $|f(A,B)|=O_f(n)$ for arbitrarily large sets $|A|=|B|=n$, then

$$f(x,y)=g(u(x)+v(y)) \text{ or } f(x,y)=g(u(x)v(y))$$

for univariate real polynomials g,u,v. Equivalently, for every fixed nonspecial f, the minimum of $|f(A,B)|/n$ tends to infinity with n. Pohoata's n^{2-c} example refutes the conjectured near-quadratic strengthening, not this qualitative theorem.

Why $f=x+y+(x-y)^2$ is not additive

If the outer polynomial g were linear, an additive form would have no xy term. Otherwise degree two forces g quadratic and u,v linear. Its quadratic homogeneous part would be a scalar square of one linear form. Matching $(x-y)^2$ forces that form to point in the x-y direction, so its linear part also points in that direction. But f has linear part x+y, a contradiction.

Why it is not multiplicative

Degree comparison forces the outer, x-side, and y-side univariate polynomials all to be linear. The quadratic homogeneous part would then be a scalar multiple of xy, whereas f has $x^2-2xy+y^2$.

Global versus local compression

Mechanism	What becomes small	Status for the fixed f
Global group form	u(A)+v(B) or u(A)v(B)	Provably absent
Local arithmetic model	Reduced image of f_Q modulo split primes	Present and amplified

What this means
Local arithmetic compression is a rigorous alternative mechanism for this example, but it is not claimed to classify every nonspecial polynomial or every source of small images.

13. Modular stress test for h(x,y)=xy(x+y)

Cycle 1's obstruction might still have been locally sparse. The finite-field test shows that this hope almost completely fails.

Homogeneous reduction

For $x \neq 0$, write $y=rx$. Then

$$h(x,rx)=x^3r(1+r).$$

If $p \equiv 2 \bmod 3$, the cube map on F_p^* is bijective. Since $h(1,1)=2$, scaling $(1,1)$ by a suitable cube root produces every nonzero value. Thus h is surjective.

The case $p \equiv 1 \bmod 3$

Let χ be a multiplicative character of order three. The number N_j of ratios r for which $r(1+r)$ lies in a prescribed cube coset satisfies

$$N_j \geq (p-2-2p^{1/2})/3.$$

[BACKGROUND] This follows by expanding the indicator of a cube coset and using the standard Jacobi-sum identity $|\Sigma_r \chi(r(r+1))|=p^{1/2}$.

The lower bound is positive for every prime $p \equiv 1 \bmod 3$ with $p \geq 13$. Hence $r(1+r)$ hits every cube coset, and h is again surjective. Direct calculation handles the remaining primes.

$$|h(F_p,F_p)| = 1 \ (p=2),\ 5 \ (p=7),\ p \text{ otherwise.}$$

For $p=7$ the image is $\{0,1,2,5,6\}$, of density $5/7$.

RATE AUDIT

The odd local saving exists only at $p=7$. Pohoata's tower theorem does not assert that 7 belongs to its split-prime set. Even if 7 split completely, the generic degree-three threshold contains the extra factor X, and the single density 5/7 does not beat the resulting archimedean and discriminant costs.

What this means

The polynomial defeats both tested routes: a geometric progression gives an almost quadratic real image, while almost every finite-field image is full. Rank-two mixing remains relevant for h specifically.

14. Comparison of the three source architectures

Feature	Unit distance	Sum-product A=GP	Polynomial local-to-global
Carrier	Coset window in $D^{-1}O_K$ inside C^f	A=GP with P an ordinary box and G a log-unit box	$P=B_K(X)$, then one real embedding
Arithmetic amplifier	Binary ideal choices from split primes	Rank-(d-1) logarithmic unit lattice	Local residue restrictions at split primes
Role of split primes	Create conjugate prime-ideal pairs and many norm-one directions	No split-prime step in the final proof	Create d residue-field coordinates per rational prime
Local structural gain	$\|U\|\geq e^{Yf}$ selected translations	$\|GG\|/\|G\|^2\leq e^{-sd}$	Allowed residue density ρ^d
Archimedean control	Exact modulus one; product-disc overlap	$\|\sigma_i(g)\|\leq e^{Y}$; additive box	$f_Q(P,P)\subset B_K(6X^2)$
Arithmetic costs	Class number, denominator, packing, root discriminant	Discriminant, regulator, distortion, directness	Discriminant D, modulus Q, coefficient/output scale
High-dimensional step	$e^{Yf/2}$ edges versus $n\leq e^{Bf}$	e^{-sd} saving versus $n\leq e^{Bd}$	$\theta^d n^2$ with $\theta<1$

Inheritance map

- From unit distance: the bounded-root-discriminant tower with a fixed infinite split-prime set, all embeddings, injective projection, and exponential-rate conversion.
- From sum-product: the symmetric additive Minkowski box, its discriminant-sensitive lattice count, and the idea that fixed per-degree gains become powers.
- New for the polynomial problem: a tailored reduced polynomial image, CRT density, uniform residue-class packing, and fixed-polynomial rescaling.

[ABSTRACTED PRINCIPLE] The uses of split primes are not identical. Unit distance turns ideal-factorization branches into many directions; polynomial compression turns quotient-ring coordinates into a sparse output support.

15. Updated mechanism library

ID	Principle	Quantitative form	Classification
P1	Factor-through compression	Exact small intermediate support with multiplicity audit	VALID WITH HYPOTHESES
P2	Bounded-fiber coordinate compression	Record domains and every fiber cost	VALID WITH HYPOTHESES
P3	Archimedean container	Bound conjugates and count lattice points	VALID WITH HYPOTHESES; TOO WEAK ALONE
P4	Local residue compression	$\lvert\Omega\rvert/R^{d} \leq e^{-sd}$	VALID WITH PER-CLASS PACKING
P5	Split-prime amplification	$q_p \to q_p^{d}$ under complete splitting	VALID WITH HYPOTHESES
P6	Fixed-polynomial rescaling	Affine bijections preserve input and output cardinality	VALID WITH HYPOTHESES
P7	Signed high-dimensional conversion	$2a-c+s>0$	VALID
P8	Collision equation	Necessary diagnostic; separate support certificate required	VALID AS DIAGNOSTIC; TOO WEAK ALONE

Main Cycle 2 certificate
large carrier + sparse local image + complete splitting + uniform per-class packing + controlled fibers + $2a-c+s>0$ implies a small global polynomial image.

16. Failure library and adversarial audit

ID	Rejected move	Precise failure	Mutation
F6	Multiply residue density by total box size	No equidistribution; all points may lie in allowed classes	Use worst-case per-class packing
F7	Any prime gives d restrictions	Number of prime ideals above p may be much smaller than d	Require complete splitting
F8	Local sparsity alone	Height, modulus, and discriminant can erase the gain	Test $\Theta<1$
F9	Let Q grow with $\|A\|$	Changes the target polynomial	Fix Q, then rescale to one fixed f
F10	$h=xy(x+y)$ is locally sparse	Full image for every odd p except 7	Retain as obstruction; prime powers remain open

Fourteen-point audit summary

Audit cluster	Result	Evidence
CRT and independence	PASS	Correct ideals; complete splitting gives d degree-one factors
Exact local density	PASS	$(p+1)/(2p)$, including zero
Several primes	PASS	Pairwise comaximal factors; densities multiply
Coefficient and Q growth	PASS for source	Q fixed; $X\geq Q$ gives $6X^2$ output radius
Discriminant cost	PASS	Included in $\theta=13D\rho$
Full-support logic	PASS	Uniform per-class packing, not heuristic density
Fixed polynomial	PASS	Two scalar bijections, fiber one
Elekes-Ronyai form	PASS	Direct degree and homogeneous-part proof
High-dimensional sign	PASS	$\theta<1$ before d tends to infinity
Generality	LIMITED	Candidate supply beyond the quadratic example remains open

17. Open lemmas

ID	Open statement	Why it matters
O1	Sparse local value-set criterion	Recognize fixed F with $\lvert F(F_p,F_p)\rvert \le (1-\eta)p$ for useful primes
O2	Removable arithmetic deformation	Classify F_Q with sparse reductions and affine rescaling to one F
O3	Higher-degree threshold	Realize $\rho D(2HX^{m-2}+R/X^2)<1$ for $m>2$
O4	Prescribed split-prime compatibility	Place candidate-specific primes in a suitable bounded-root-discriminant tower
O5	Prime-power amplification	Control value sets and class packing modulo p^k
O6	Rank-two correlated sums	Retained for polynomials with full local images
O7	Carriers beyond symmetric boxes	Retain computable uniform per-class exponent u

[OPEN] The exact pre-Phase-C bottleneck is to find, or rule out, new fixed polynomials admitting both a sufficiently sparse local deformation and a rate-compatible rescaling. Without this supply step, random construction search has no validated filter.

18. Final bottleneck and recommendation

	Question	Answer
Q1	Is rank-two mixing still the main bottleneck?	Not universally. Local residue compression bypasses it for Pohoata's polynomial, but it remains relevant when local images are full.
Q2	Does congruence compression rigorously bypass global factorization failure?	Yes, under complete splitting, uniform class packing, controlled fibers, and a positive signed rate.
Q3	Inherited from unit distance?	The split-prime tower, all-embedding control, injective projection, and exponential-rate architecture.
Q4	Inherited from high-degree sum-product?	The symmetric additive Minkowski box, discriminant-sensitive counting, and degree-to-power conversion.
Q5	Genuinely new?	Sparse reduced polynomial images, CRT support density, per-class output packing, and fixed-polynomial rescaling.
Q6	Strongest reusable abstraction?	Uniform residue-class packing plus 2a-c+s>0.
Q7	Exact remaining bottleneck?	Supply and recognize rate-compatible removable local deformations for new fixed polynomials.
Q8	Next action?	Targeted literature review, not unrestricted Phase C.

Recommendation: targeted literature review
Review finite-field polynomial value sets, exceptional and sparse-image polynomials, character-sum methods, prime-power lifting, and prescribed split-prime towers. The framework is now strong enough to specify what to search for, but not yet strong enough to justify unrestricted Phase C construction experiments.

[ABSTRACTED PRINCIPLE] Cycle 2 succeeds: local arithmetic sparsity, repeated through completely split primes and combined with uniform global height control, is a rigorous reusable mechanism for polynomial-image compression.

References

1. C. Pohoata, Split primes and the Elekes-Ronyai problem, arXiv:2606.13619v1, 11 June 2026. https://arxiv.org/abs/2606.13619

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3. O. E. Raz, M. Sharir, and J. Solymosi, Polynomials vanishing on grids: the Elekes-Ronyai problem revisited, American Journal of Mathematics 138 (2016), 1029-1065. Author manuscript used to verify the precise restatement of the original theorem.

4. OpenAI, Planar Point Sets with Many Unit Distances, supplied proof manuscript, especially Proposition 2.2, Theorem 2.3, and Lemmas 2.4-2.6.

5. N. Alon et al., Remarks on the Disproof of the Unit Distance Conjecture, supplied human-verified remarks manuscript.

6. Phase A - Reproducing the High-Degree Sum-Product Construction, final frozen project document, 26 August 2026.

7. Phase B - From Unit Distance and Sum-Product to Polynomial Image Compression, Cycle 1 project document, 26 August 2026.

Source discrepancy check

No normalization discrepancy was found: the paper uses f_Q(x,y)=Q(x+y)+(x-y)^2 and rescales to f(x,y)=x+y+(x-y)^2 exactly as anticipated. The substantive correction concerns the abstract lemma: local density requires a uniform per-class packing bound before it can be converted into global support compression.